Celanese
The chemistry inside innovation™

Hytrel® 5556

THERMOPLASTIC POLYESTER ELASTOMER

Common features of Hytrel® thermoplastic polyester elastomer include mechanical and physical properties such as exceptional toughness and resilience, high resistance to creep, impact and flex fatigue, flexibility at low temperatures and good retention of properties at elevated temperatures. In addition, it resists many industrial chemicals, oils and solvents. Special grades include heat stabilised, flame retardant, food contact compliant, blow molding and extrusion grades. Concentrates offered include black pigments, UV protection additives, heat stabilisers, and flame retardants. Hytrel® thermoplastic polyester elastomer is plasticiser free.

The good melt stability of Hytrel® thermoplastic polyester elastomer normally enables the recycling of properly handled production waste. If recycling is not possible, we recommend, as the preferred option, incineration with energy recovery (-24 kJ/g of base polymer) in appropriately equipped installations.
For disposal, local regulations have to be observed.

Hytrel® thermoplastic polyester elastomer typically is used in demanding applications in the automotive, fluid power, electrical/electronic, consumer goods, appliance and power tool, sporting goods, furniture, industrial and off-road transportation/equipment industry.

Hytrel® 5556 is a medium modulus Hytrel® grade with nominal durometer hardness of 55D. It contains non-discoloring stabilizer. It can be processed by many conventional thermoplastic processing techniques like injection moulding and extrusion.

Typical applications: Hose and tubing, wire and cable, film and sheeting, belting.

Product information

Resin Identification	TPC-ET	ISO 1043
Part Marking Code	>TPC-ET<	ISO 11469
TGA curve	available	ISO 11358-1

Rheological properties

Melt volume-flow rate	7	cm^3/10min	ISO 1133
Temperature	220	°C	
Load	2.16	kg	
Melt mass-flow rate	7.8	g/10min	ISO 1133
Melt mass-flow rate, Temperature	220	°C	
Melt mass-flow rate, Load	2.16	kg	
Molding shrinkage, parallel	1.4	%	ISO 294-4, 2577
Molding shrinkage, normal	1.4	%	ISO 294-4, 2577

Typical mechanical properties

Shore D hardness, 15s	51		ISO 48-4 / ISO 868
Shore D hardness, max	55		ISO 868
Tensile modulus	180	MPa	ISO 527-1/-2
Tensile stress at yield	15	MPa	ISO 527-1/-2
Tensile strain at yield	40	%	ISO 527-1/-2
Stress at 5% strain	6.9	MPa	ISO 527-1/-2

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Stress at 10% strain	11	MPa	ISO 527-1/-2
Tensile stress at 50% strain, 1BA	14.5	MPa	ISO 527-1/-2
Tensile stress at 100% strain	16	MPa	ISO 527-1/-2
Tensile stress at break	40	MPa	ISO 527-1/-2
Nominal strain at break	600	%	ISO 527-1/-2
Tensile strain at break	>300	%	ISO 527-1/-2
Flexural modulus	190	MPa	ISO 178
Tensile creep modulus, 1h	170	MPa	ISO 899-1
Tensile creep modulus, 1000h	133	MPa	ISO 899-1
Charpy impact strength, 23°C	N	kJ/m²	ISO 179/1eU
Charpy impact strength, -30°C	N	kJ/m²	ISO 179/1eU
Charpy notched impact strength, 23°C	N	kJ/m²	ISO 179/1eA
Charpy notched impact strength, -30°C	145[P]	kJ/m²	ISO 179/1eA
Charpy notched impact strength, -40°C	145	kJ/m²	ISO 179/1eA
Tensile notched impact strength, 23°C	320	kJ/m²	ISO 8256/1
Puncture - maximum force, 23°C	2400	N	ISO 6603-2
Puncture - maximum force, -30°C	3700	N	ISO 6603-2
Puncture energy, 23°C	27	J	ISO 6603-2
Puncture energy, -30°C	43	J	ISO 6603-2
Izod notched impact strength, 23°C	N	kJ/m²	ISO 180/1A
Izod notched impact strength, -40°C	N	kJ/m²	ISO 180/1A
Brittleness temperature	-98	°C	ISO 974
Tear strength, parallel	140	kN/m	ISO 34-1
Tear strength, normal	130	kN/m	ISO 34-1
Abrasion resistance	120	mm³	ISO 4649

[P]: Partial Break

Thermal properties

Melting temperature, 10°C/min	201	°C	ISO 11357-1/-3
Glass transition temperature, 1 Hz	-27	°C	ISO 6721
Temperature of deflection under load, 1.8 MPa	45	°C	ISO 75-1/-2
Temperature of deflection under load, 0.45 MPa	70	°C	ISO 75-1/-2
Vicat softening temperature, 50°C/h 50N	75	°C	ISO 306
Vicat softening temperature, 50°C/h 10N	180	°C	ISO 306
Coefficient of linear thermal expansion (CLTE), parallel	180	E-6/K	ISO 11359-1/-2
Coefficient of linear thermal expansion (CLTE), parallel, -40-23°C	160	E-6/K	ISO 11359-1/-2
Coefficient of linear thermal expansion (CLTE), normal	180	E-6/K	ISO 11359-1/-2
Coefficient of linear thermal expansion (CLTE), normal, -40-23°C	174	E-6/K	ISO 11359-1/-2
Thermal conductivity of melt	0.16	W/(m K)	ISO 22007-2
Effective thermal diffusivity, flow	7E-8	m²/s	ISO 22007-4
Specific heat capacity of melt	2110	J/(kg K)	ISO 22007-4
RTI, electrical, 0.75mm	85	°C	UL 746B
RTI, electrical, 1.5mm	85	°C	UL 746B
RTI, electrical, 3.0mm	85	°C	UL 746B
RTI, impact, 0.75mm	50	°C	UL 746B

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RTI, impact, 1.5mm	85	°C	UL 746B
RTI, impact, 3.0mm	85	°C	UL 746B
RTI, strength, 0.75mm	50	°C	UL 746B
RTI, strength, 1.5mm	75	°C	UL 746B
RTI, strength, 3.0mm	80	°C	UL 746B

Flammability

Burning Behav. at 1.5mm nom. thickn.	HB	class	IEC 60695-11-10
Thickness tested	1.5	mm	IEC 60695-11-10
UL recognition	yes		UL 94
Burning Behav. at thickness h	HB	class	IEC 60695-11-10
Thickness tested	3	mm	IEC 60695-11-10
UL recognition	yes		UL 94
Oxygen index	20	%	ISO 4589-1/-2
Glow Wire Flammability Index, 2.0mm	700[DS]	°C	IEC 60695-2-12
Glow Wire Ignition Temperature, 2.0mm	675[DS]	°C	IEC 60695-2-13
Glow Wire Temperature, No Flame, 2mm	650[DS]	°C	IEC 60335-1
FMVSS Class	SE		ISO 3795 (FMVSS 302)

[DS]: Derived from similar grade

Electrical properties

Relative permittivity, 100Hz	4.8		IEC 62631-2-1
Relative permittivity, 1MHz	4.4		IEC 62631-2-1
Dissipation factor, 100Hz	90	E-4	IEC 62631-2-1
Dissipation factor, 1MHz	375	E-4	IEC 62631-2-1
Volume resistivity	4E11	Ohm.m	IEC 62631-3-1
Surface resistivity	>1E15	Ohm	IEC 62631-3-2
Electric strength	19	kV/mm	IEC 60243-1
Comparative tracking index	600		IEC 60112

Physical/Other properties

Humidity absorption, 2mm	0.2	%	Sim. to ISO 62
Water absorption, 2mm	0.6	%	Sim. to ISO 62
Water absorption, Immersion 24h	0.6	%	Sim. to ISO 62
Density	1190	kg/m³	ISO 1183
Density of melt	1030	kg/m³	

Film properties

WVTR, 23°C/85%r.h.	300	g/(m²*d)	DIS 15106-1/-2
Oxygen transmission rate, 23°C/85%r.h.	6000	cm³/(m²*d*bar)	DIS 15105-1/-2

Emissions

Light stability delta l	3		DIN 53236
Light stability delta a	-2		DIN 53236
Light stability delta b	13		DIN 53236
Light stability delta E	14		DIN 53236
Emission of organic compounds	8.5	µgC/g	VDA 277

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Odor test	5	class	VDA 270
Fogging, G-value (condensate)	0.1	mg	ISO 6452

Injection

Drying Recommended	yes	
Drying Temperature	100	°C
Drying Time, Dehumidified Dryer	2 - 3	h
Processing Moisture Content	≤0.08	%
Melt Temperature Optimum	235	°C
Min. melt temperature	220	°C
Max. melt temperature	250	°C
Mold Temperature Optimum	50	°C
Min. mold temperature	45	°C
Max. mold temperature	55	°C
Hold Pressure Time (h is the max. wall thickness of the part in mm)	refer to molding guide	s
Ejection temperature	122	°C

Extrusion

Drying Temperature	90 - 110	°C
Drying Time, Dehumidified Dryer	2 - 3	h
Processing Moisture Content	≤0.06	%
Melt Temperature Optimum	225	°C
Melt Temperature Range	220 - 235	°C

Automotive

OEM	STANDARD	ADDITIONAL INFORMATION
Bosch	N28 BN34-OX067	
General Motors	GMW17327P-TPC-ET-Type 3M2	
Mercedes-Benz	DBL5562 TPC	
Stellantis - Chrysler	MS-DB-448 / CPN-3042	Natural
VW Group	VW 50123 TPC-ET	

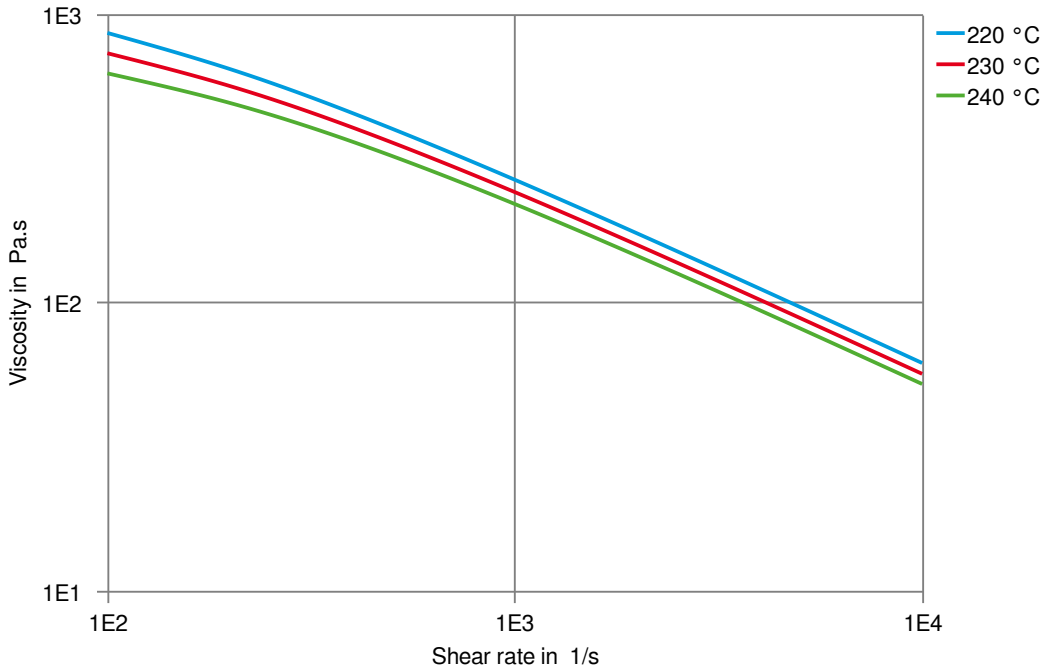
Characteristics

Processing	Injection Molding, Film Extrusion, Extrusion, Sheet Extrusion, Other Extrusion, Coating, Casting, Thermoforming
Special characteristics	Light stabilized or stable to light

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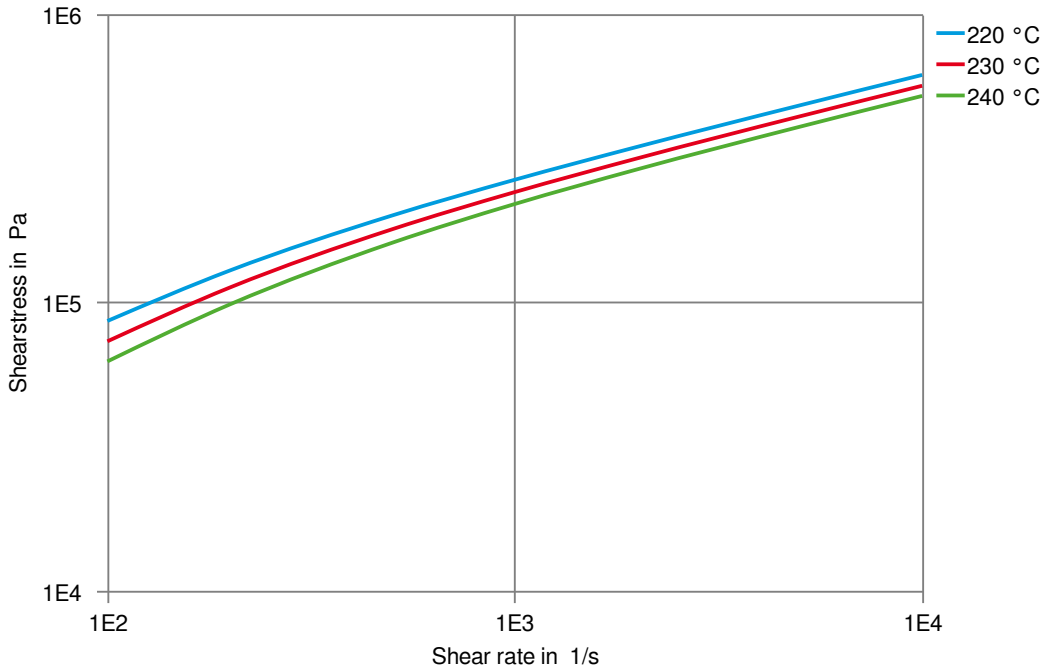
Viscosity-shear rate



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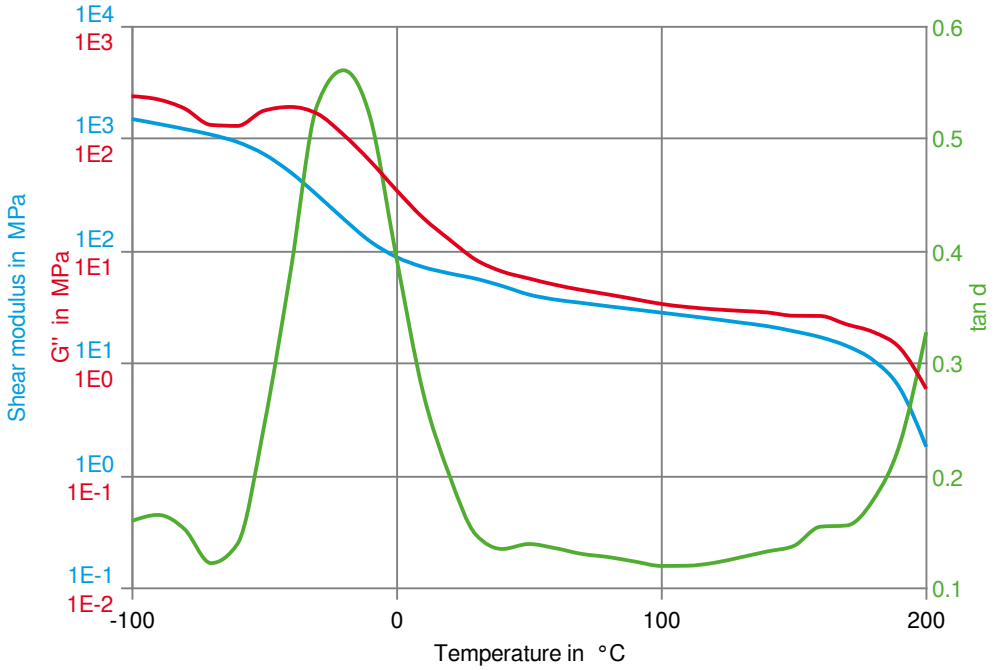
Shearstress-shear rate



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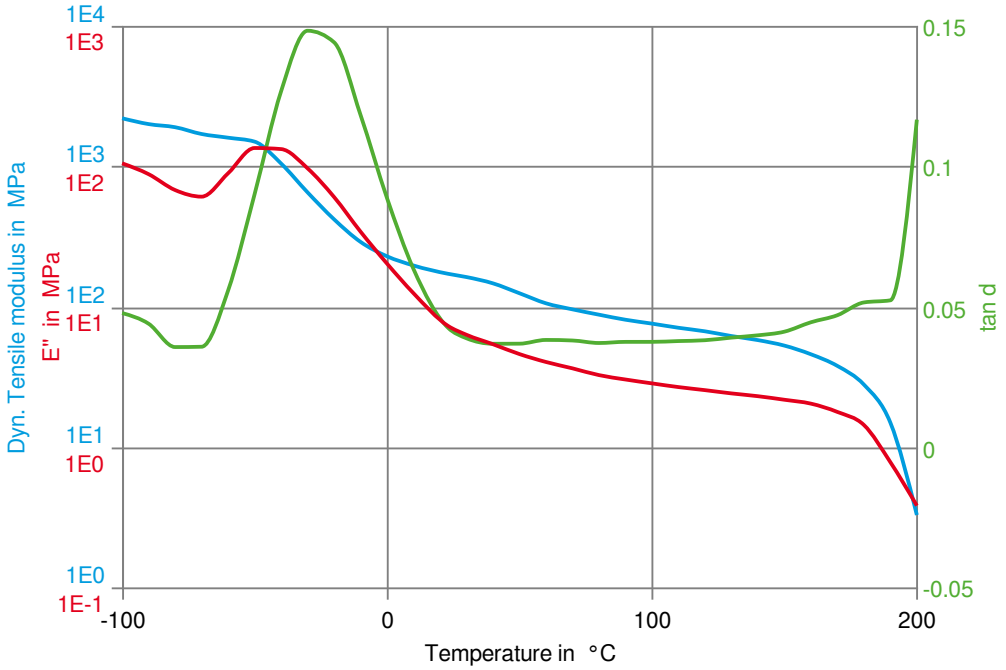
Dynamic Shear modulus-temperature



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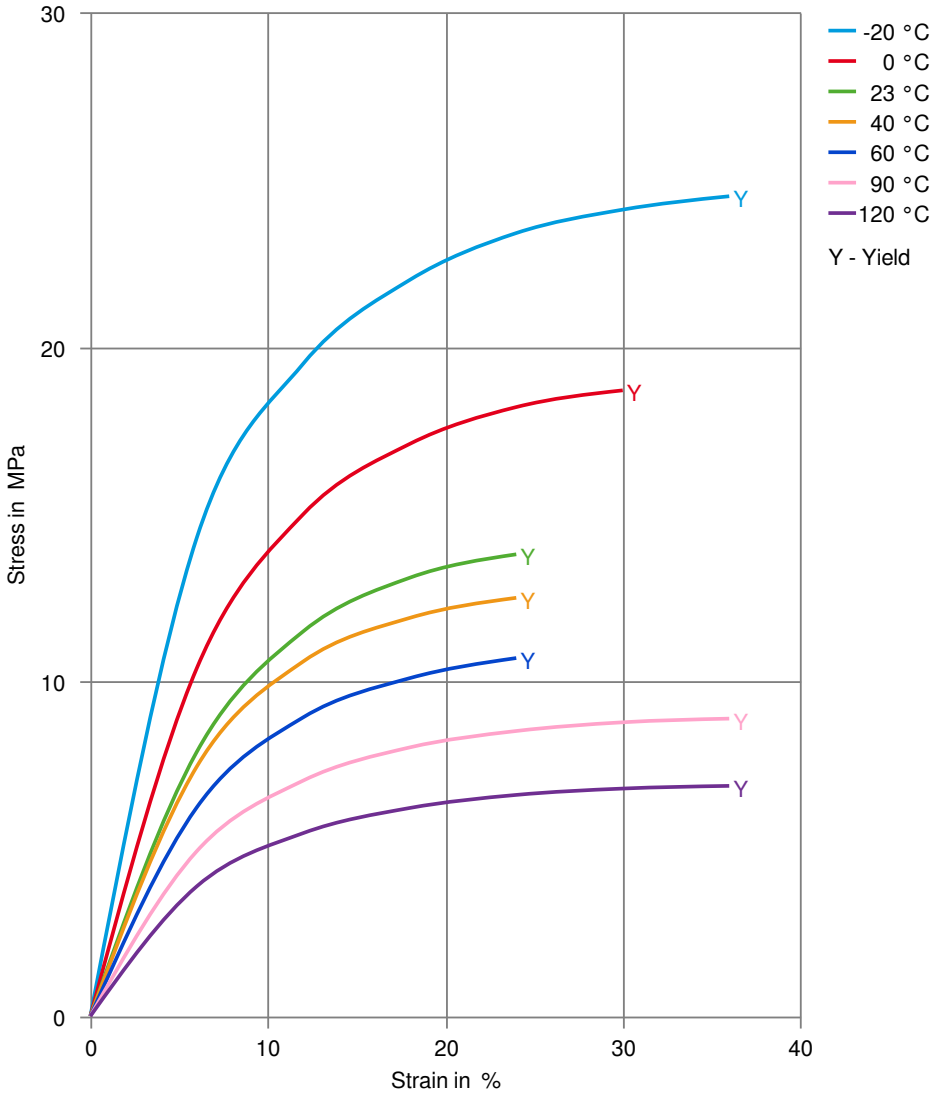
Dynamic Tensile modulus-temperature



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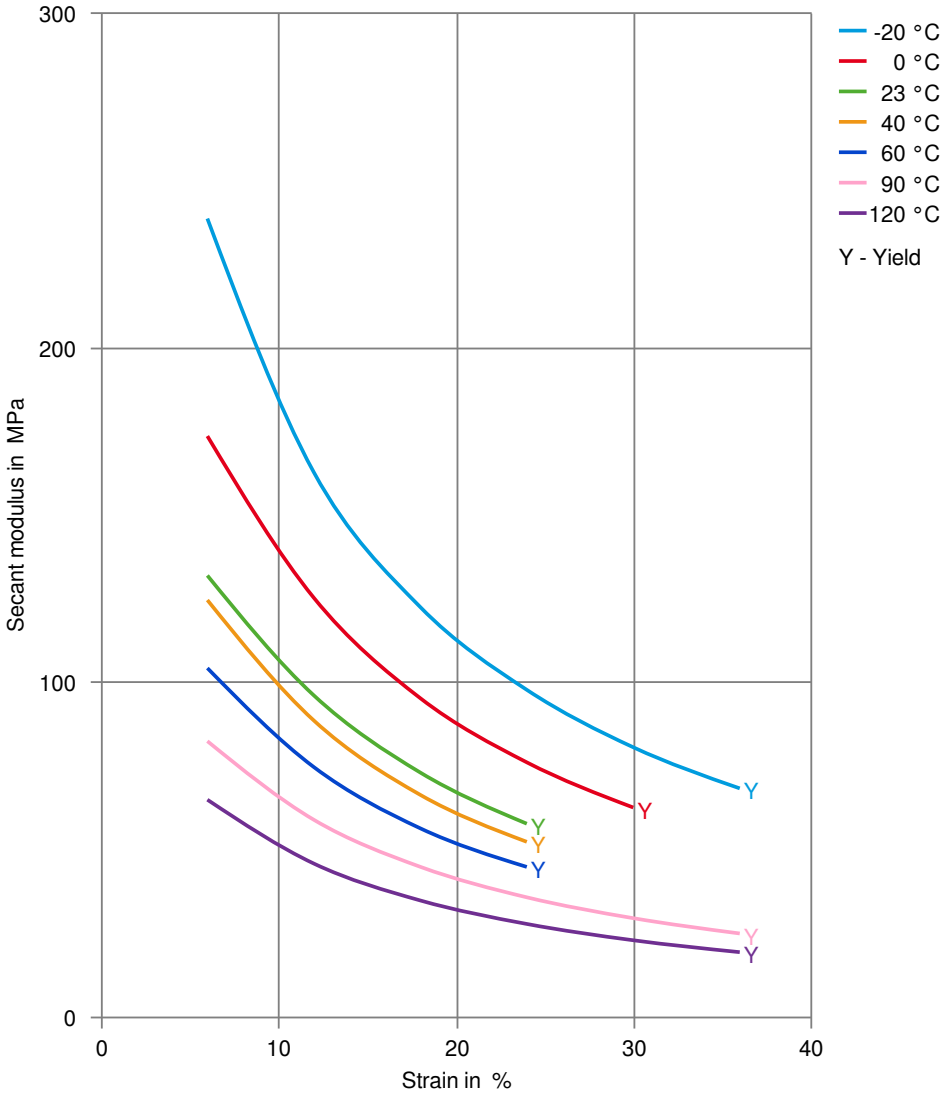
Stress-strain



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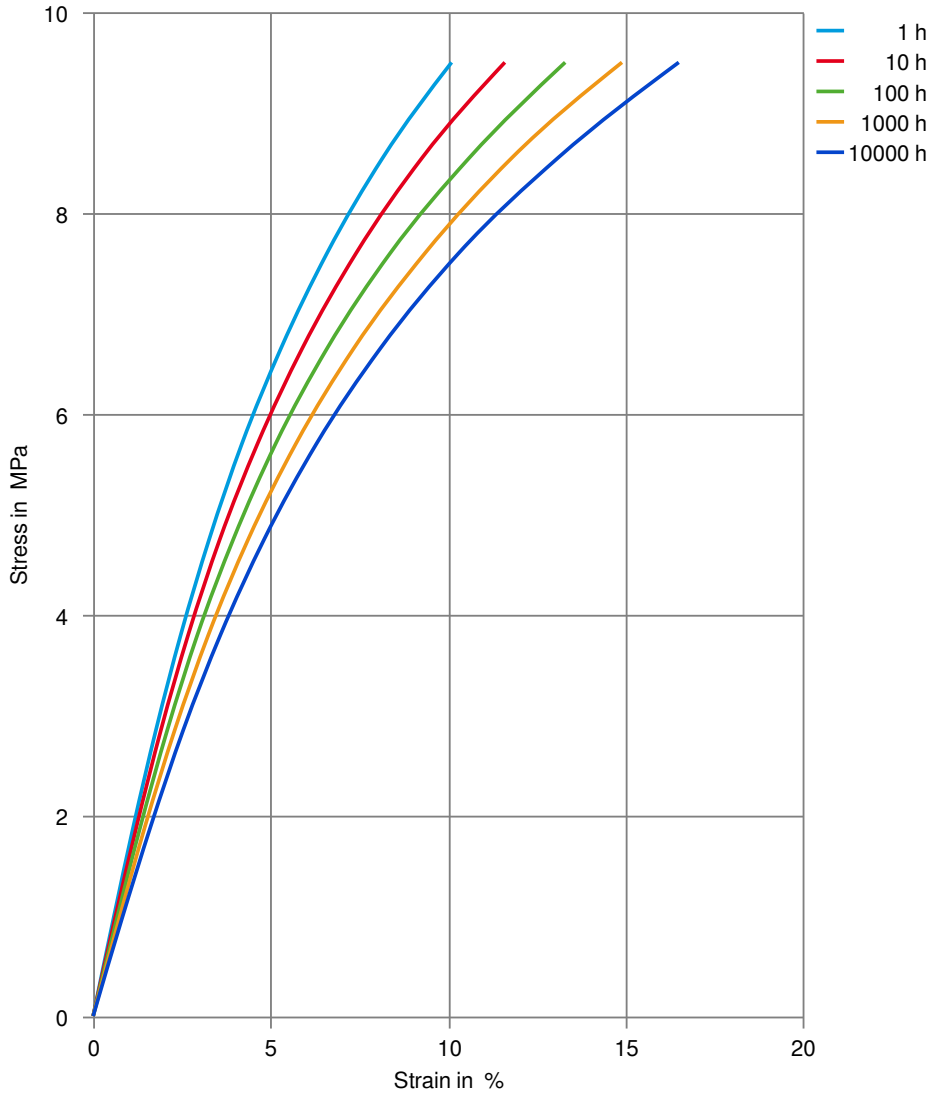
Secant modulus-strain



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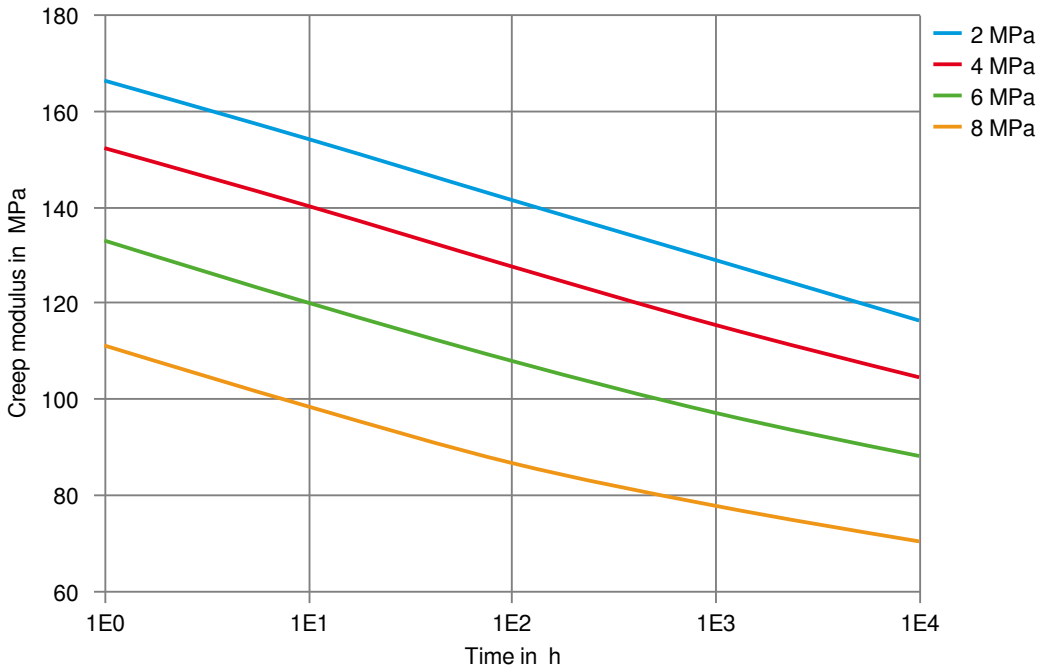
Stress-strain (isochronous) 23 °C



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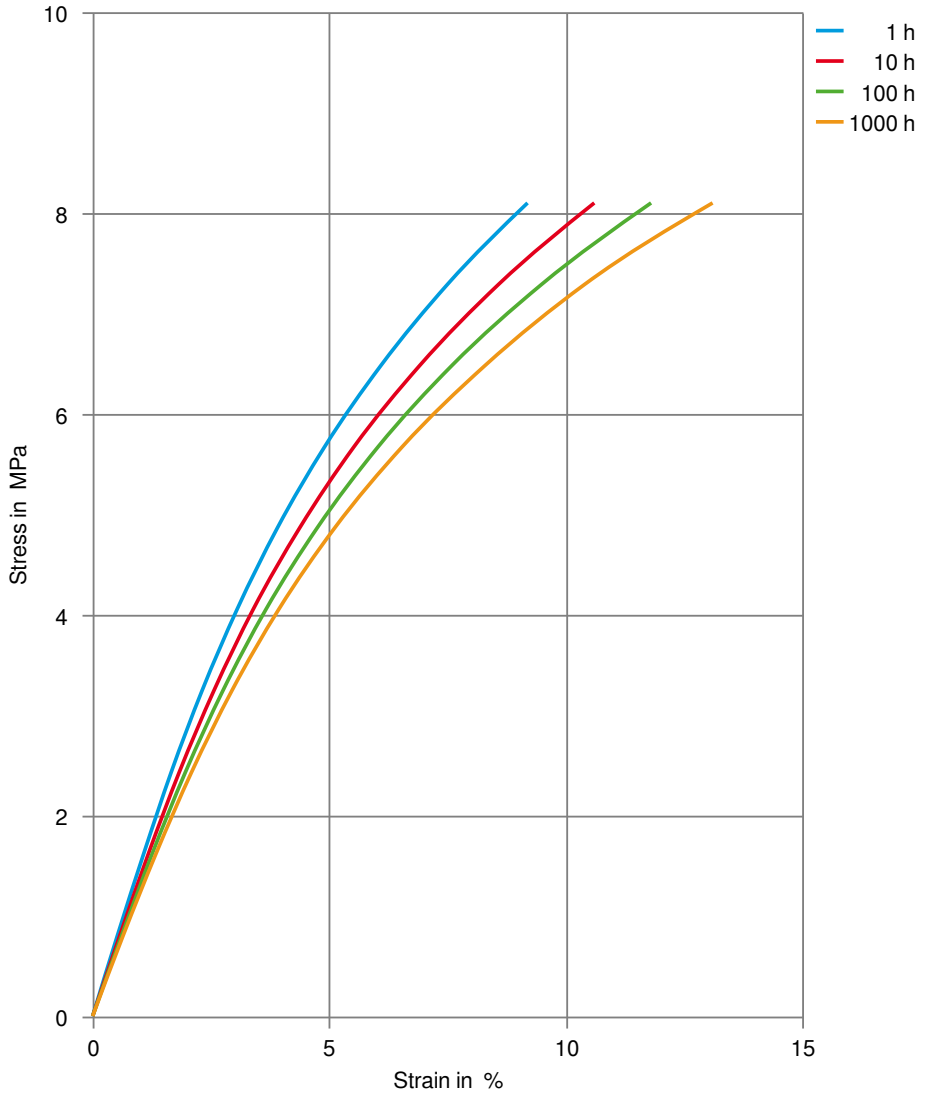
Creep modulus-time 23 °C



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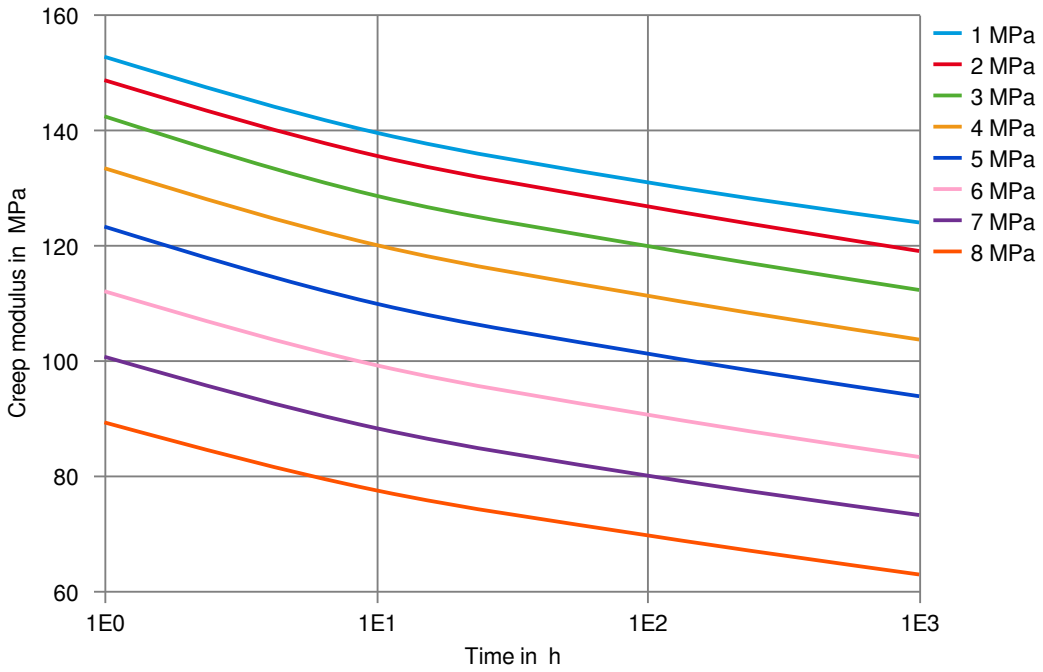
Stress-strain (isochronous) 40 °C



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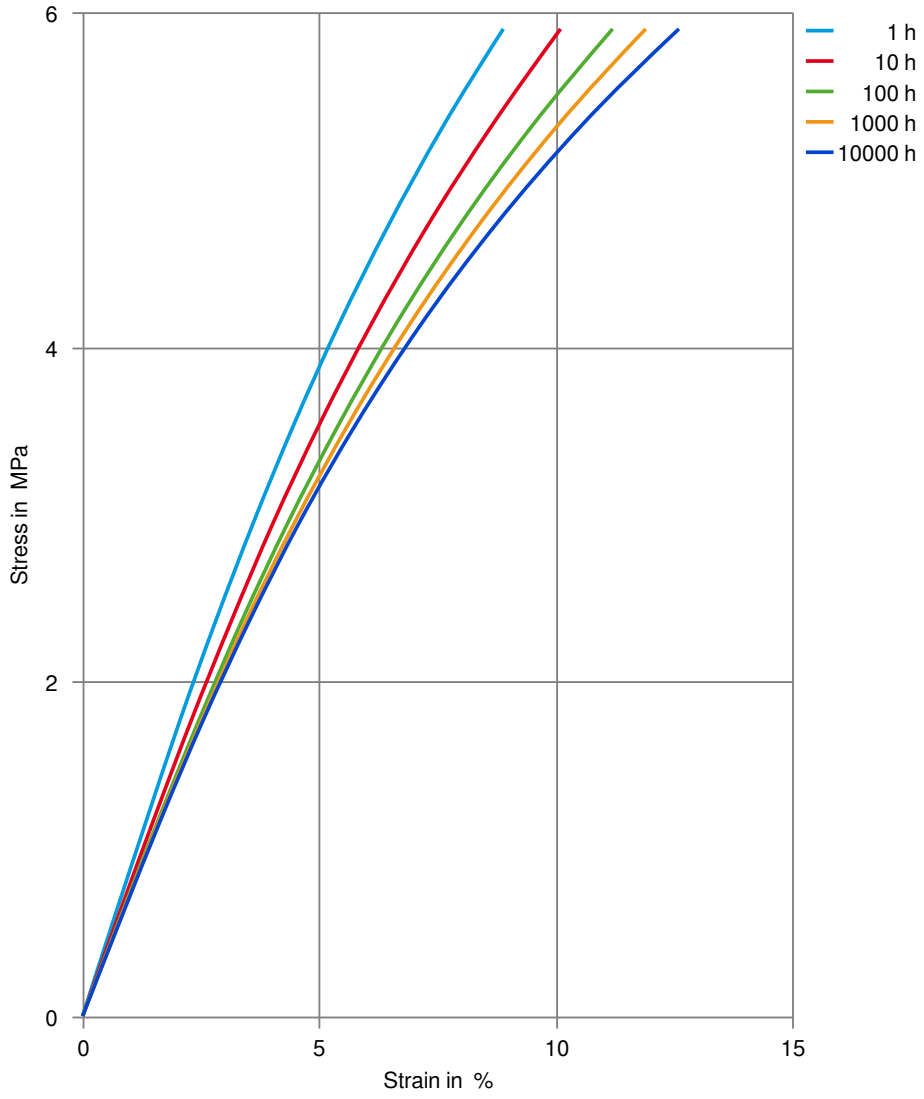
Creep modulus-time 40 °C



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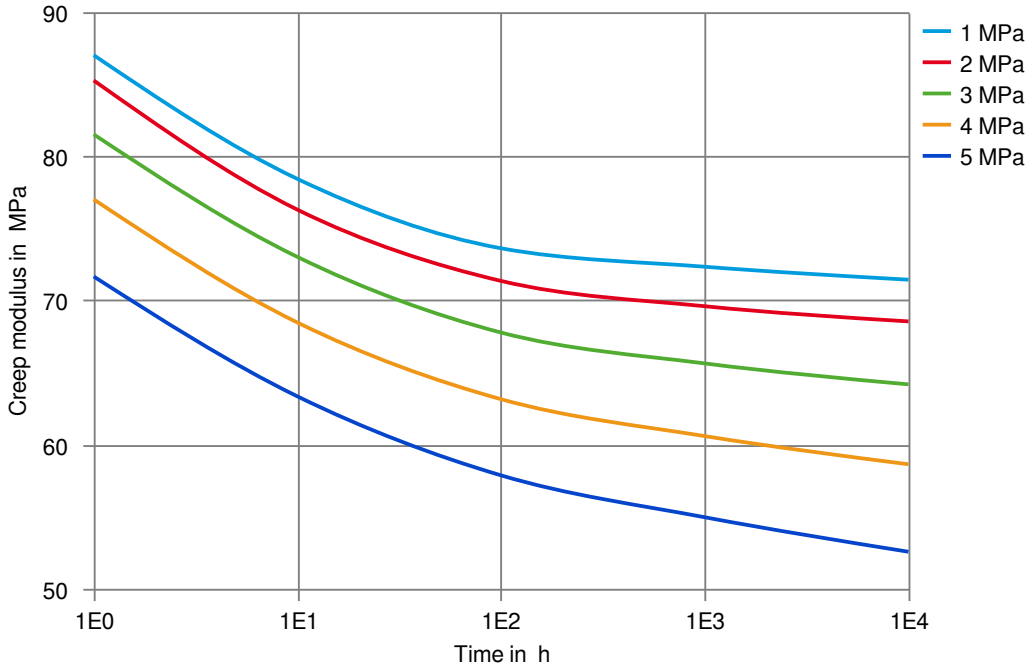
Stress-strain (isochronous) 80 °C



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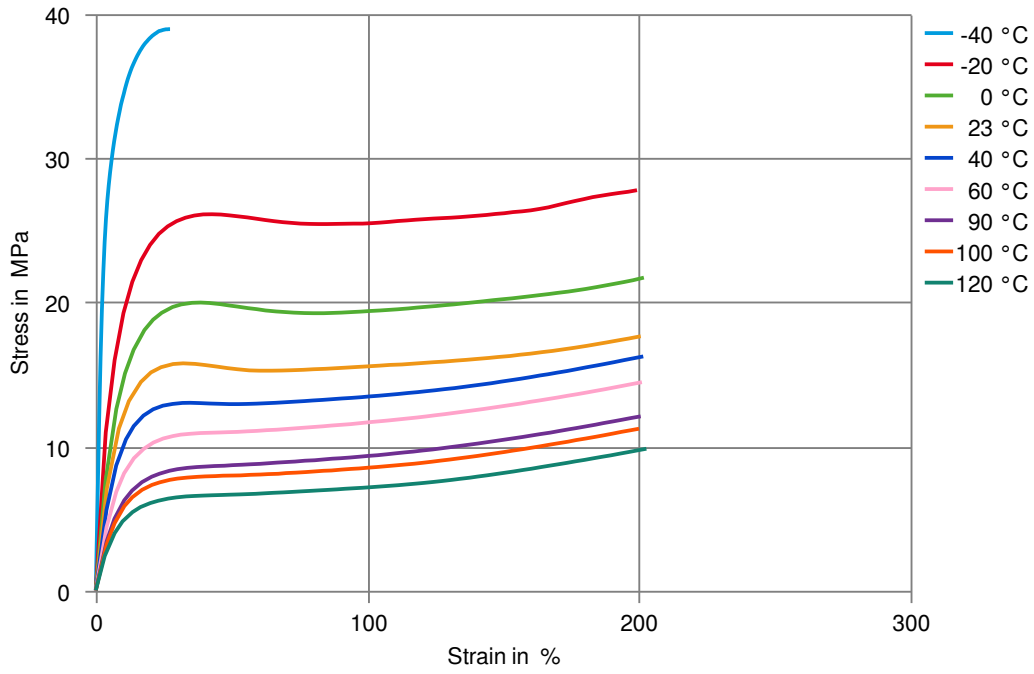
Creep modulus-time 80 °C



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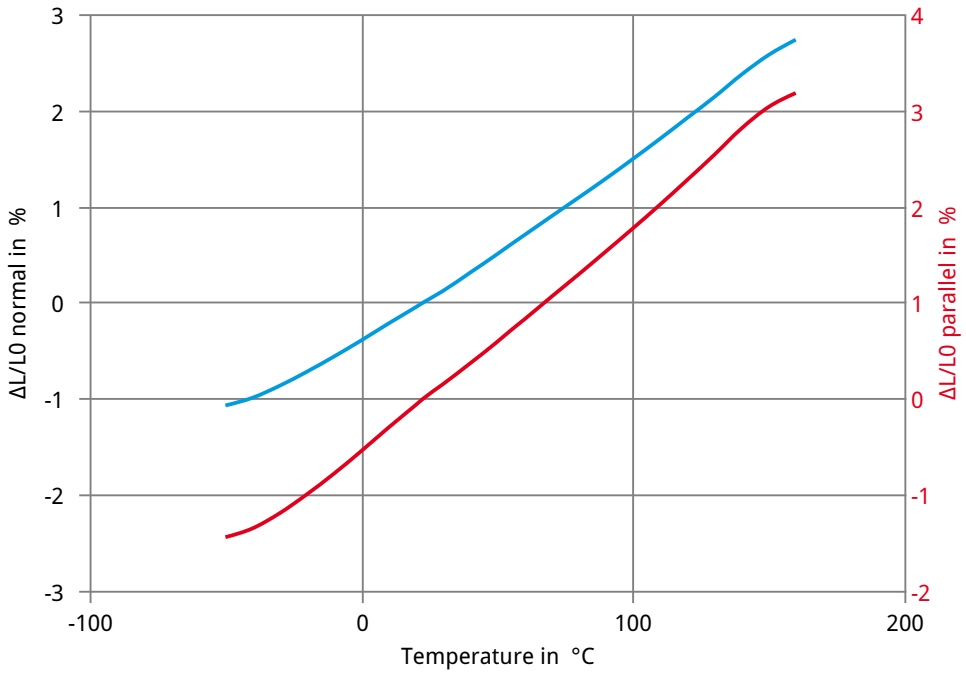
Stress-Strain (Flexible Materials)



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Thermal expansion



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Chemical Media Resistance

Acids

- ✔ Acetic Acid (5% by mass), 23°C
- ✔ Citric Acid solution (10% by mass), 23°C
- ✔ Lactic Acid (10% by mass), 23°C
- ✘ Hydrochloric Acid (36% by mass), 23°C
- ✘ Nitric Acid (40% by mass), 23°C
- ✘ Sulfuric Acid (38% by mass), 23°C
- ✔ Sulfuric Acid (5% by mass), 23°C
- ✘ Chromic Acid solution (40% by mass), 23°C

Bases

- ✔ Sodium Hydroxide solution (35% by mass), 23°C
- ✔ Sodium Hydroxide solution (1% by mass), 23°C
- ✔ Ammonium Hydroxide solution (10% by mass), 23°C

Alcohols

- ✔ Isopropyl alcohol, 23°C
- ✔ Methanol, 23°C
- ✔ Ethanol, 23°C

Hydrocarbons

- ✔ n-Hexane, 23°C
- ✔ Toluene, 23°C
- ✔ iso-Octane, 23°C

Ketones

- ✘ Acetone, 23°C

Ethers

- ✘ Diethyl ether, 23°C

Mineral oils

- ✔ SAE 10W40 multigrade motor oil, 23°C
- ✘ SAE 10W40 multigrade motor oil, 130°C
- ✘ SAE 80/90 hypoid-gear oil, 130°C
- ✔ Insulating Oil, 23°C
- ✘ Motor oil OS206 304 Ref.Eng.Oil, ISP, 135°C
- ✘ Automatic hypoid-gear oil Shell Donax TX, 135°C
- ✘ Hydraulic oil Pentosin CHF 202, 125°C

Standard Fuels

- ✘ ISO 1817 Liquid 1 - E5, 60°C
- ✘ ISO 1817 Liquid 2 - M15E4, 60°C
- ✘ ISO 1817 Liquid 3 - M3E7, 60°C
- ✘ ISO 1817 Liquid 4 - M15, 60°C
- ✔ Standard fuel without alcohol (pref. ISO 1817 Liquid C), 23°C
- ✔ Standard fuel with alcohol (pref. ISO 1817 Liquid 4), 23°C
- ✔ Diesel fuel (pref. ISO 1817 Liquid F), 23°C
- ✔ Diesel fuel (pref. ISO 1817 Liquid F), 90°C
- ✘ Diesel fuel (pref. ISO 1817 Liquid F), >90°C

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Salt solutions

- ✔ Sodium Chloride solution (10% by mass), 23°C
- ✘ Sodium Hypochlorite solution (10% by mass), 23°C
- ✔ Sodium Carbonate solution (20% by mass), 23°C
- ✔ Sodium Carbonate solution (2% by mass), 23°C
- ✔ Zinc Chloride solution (50% by mass), 23°C

Other

- ✔ Ethyl Acetate, 23°C
- ✘ Hydrogen peroxide, 23°C
- ✘ DOT No. 4 Brake fluid, 130°C
- ✘ Ethylene Glycol (50% by mass) in water, 108°C
- ✔ 50% Oleic acid + 50% Olive Oil, 23°C
- ✔ Water, 23°C
- ✔ Water, 90°C
- ✔ Phenol solution (5% by mass), 23°C

Symbols used:

✔ possibly resistant
Defined as: Supplier has sufficient indication that contact with chemical can be potentially accepted under the intended use conditions and expected service life. Criteria for assessment have to be indicated (e.g. surface aspect, volume change, property change).

✘ not recommended - see explanation
Defined as: Not recommended for general use. However, short-term exposure under certain restricted conditions could be acceptable (e.g. fast cleaning with thorough rinsing, spills, wiping, vapor exposure).

Revised: 2026-08-12 Source: Celanese Materials Database

NOTICE TO USERS: Values shown are based on testing of laboratory test specimens and represent data that fall within the standard range of properties for natural material. These values alone do not represent a sufficient basis for any part design and are not intended for use in establishing maximum, minimum, or ranges of values for specification purposes. Colorants or other additives may cause significant variations in data values. Properties of molded parts can be influenced by a wide variety of factors including, but not limited to, material selection, additives, part design, processing conditions and environmental exposure. Other than those products expressly identified as medical grade (including by MT® product designation or otherwise), Celanese's products are not intended for use in medical or dental implants. Regardless of any such product designation, any determination of the suitability of a particular material and part design for any use contemplated by the users and the manner of such use is the sole responsibility of the users, who must assure themselves that the material as subsequently processed meets the needs of their particular product or use. To the best of our knowledge, the information contained in this publication is accurate; however, we do not assume any liability whatsoever for the accuracy and completeness of such information. The information contained in this publication should not be construed as a promise or guarantee of specific properties of our products. It is the sole responsibility of the users to investigate whether any existing patents are infringed by the use of the materials mentioned in this publication. Moreover, there is a need to reduce human exposure to many materials to the lowest practical limits in view of possible adverse effects. To the extent that any hazards may have been mentioned in this publication, we neither suggest nor guarantee that such hazards are the only ones that exist. We recommend that persons intending to rely on any recommendation or to use any equipment, processing technique or material mentioned in this publication should satisfy themselves that they can meet all applicable safety and health standards. We strongly recommend that users seek and adhere to the manufacturer's current instructions for handling each material they use, and entrust the handling of such material to adequately trained personnel only. Please call the telephone numbers listed for additional technical information. Call Customer Services for the appropriate Materials Safety Data Sheets (MSDS) before attempting to process our products.
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